



Data Model Design Document

Comprehensive Salesforce CRM Overhaul for Customer Operations and Service Delivery

4.1 Object Definition

To design a scalable and well-structured data architecture that represents physical infrastructure assets without causing data duplication, a combination of standard and custom objects is implemented as follows.

4.1.1 Standard Objects

- **Account:** Represents the primary customer entity or household that receives utility distribution services from GreenGrid.
- **Contact:** Stores detailed personal information, including contact numbers and communication records of individuals responsible for billing or communication.
- **Case:** Serves as the main service ticket used to record, manage, and track customer complaints from submission to resolution.

4.1.2 Custom Objects

- **Utility_Asset__c:** Represents the physical utility meter installed in the field. This object contains a unique alphanumeric identifier optimized for fast search performance within Salesforce.
- **Field_Visit__c:** Used to record field service activities, including visit history, travel details, repair actions, and technical resolutions performed by technicians.

4.2 Field Definition

4.2.1 Standard Object

To support specific customer service workflows in utility management, the standard **Case** object is customized. Instead of creating a new object, the existing **Type** field is extended with

additional values relevant to the industry. This approach improves ticket classification while maintaining system efficiency.

Standard Object: Case

- **Case Type (Type)** – *Picklist*
 - Outage: Used to identify utility distribution failures.
 - Maintenance: Used for scheduling routine maintenance and updating asset conditions to prevent potential failures.

4.2.2 Custom Object

The following fields define the essential data required to support business processes, enforce validation, and meet the targeted search performance (under 2 seconds).

Custom Object: Utility_Asset__c

- **Asset Name (Name)** – *Auto Number*
Automatically generated identifier with format MET-{0000}.
- **Meter ID (Meter_ID__c)** – *Text Area (50)*
A unique identifier used by agents for quick lookup.
Configured as Unique and External ID.
- **Asset Status (Asset_Status__c)** – *Picklist*
Indicates the operational condition of the asset.
Values: Active, Inactive, Suspended.
- **Installation Address (Installation_Address__c)** – *Text Area (255)*
Stores the exact physical location where the meter is installed.
- **Customer Account (Customer_Account__c)** – *Lookup (Account)*
Links the asset to the associated customer account.

Custom Object: Field_Visit__c

- **Visit Number (Name)** – *Auto Number*
Automatically generated identifier with format VIS-{0000}.

- **Parent Case (Parent_Case__c)** – *Lookup (Case)*
Connects the visit record to its corresponding service case.
Includes validation rules to maintain historical integrity.
- **Repair Status (Repair_Status__c)** – *Picklist*
Required field to track progress of field work.
Values: Assigned, In Progress, Resolved, Failed.
- **Technician Notes (Technician_Notes__c)** – *Long Text Area*
Used by technicians to record on-site findings directly.
- **Resolution Timestamp (Resolution_Timestamp__c)** – *Date/Time*
Automatically populated when status changes to Resolved, used for KPI tracking.

4.3 Relationship Mapping

This section describes how objects are connected within the GreenGrid Utilities Salesforce system. These relationships ensure data consistency while also supporting audit and compliance requirements.

The following outlines the relationship types, parent-child structures, and system behavior regarding deletion rules:

- **Account → Contact** (Standard Lookup)
When an Account is deleted, related Contact records remain, but the relationship field is cleared.
- **Account → Utility_Asset__c** (Custom Lookup)
Asset records remain intact and can be reassigned if the associated account changes.
- **Account → Case** (Standard Lookup)
Deletion is restricted if related Cases exist, ensuring service history is preserved.
- **Utility_Asset__c → Case** (Custom Lookup)
Service history remains available even if the asset is updated or replaced.
- **Case → Field_Visit__c** (Master-Detail / Controlled Lookup)
Deletion is restricted to maintain technical and operational records.

Relationship Explanation

- **Account to Contact**

Implemented as a standard Lookup relationship, allowing one account to be associated with multiple contacts while keeping data independent.

- **Account to Utility_Asset__c**

Uses a custom Lookup to ensure that physical assets are not dependent on account ownership changes.

- **Account to Case**

Configured with restricted deletion to preserve historical service records.

- **Utility_Asset__c to Case**

Links service tickets to assets, enabling quick access to maintenance history.

- **Case to Field_Visit__c**

Implemented as a Master-Detail relationship to ensure that field visits are always tied to a valid case and inherit its security and sharing settings.

4.4 Salesforce Implementation and Configuration

Step 1 : Creating Developer Account

- a. Go to [Developer Edition Org | Salesforce](#)
- b. On the sign up form, enter the following details :

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Sign up for your Developer Edition

A free Salesforce Platform environment with Agentforce and Data 360

First name

mecek ✓

Last name

Smbrg ✓

Job title

Student ✓

Company

SmartBridge ✓

Country/region

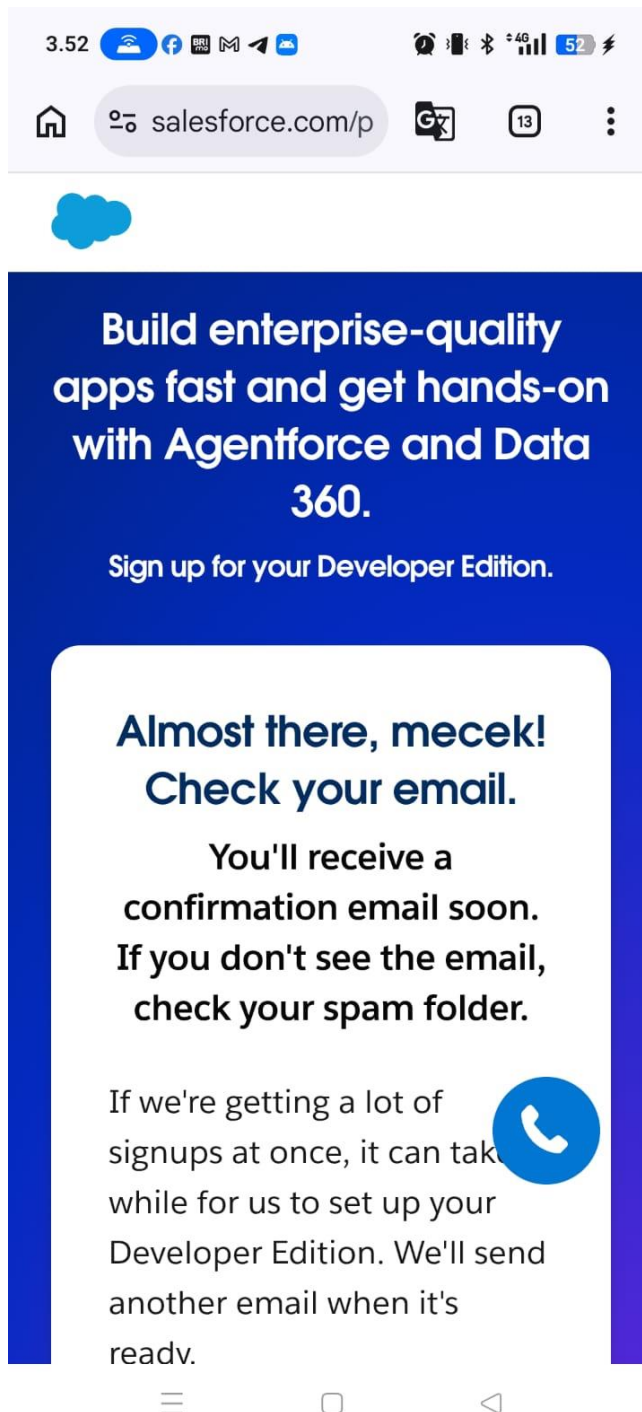
Indonesia ✓ 

Work email

mecekkristinabrsembiri... ✓

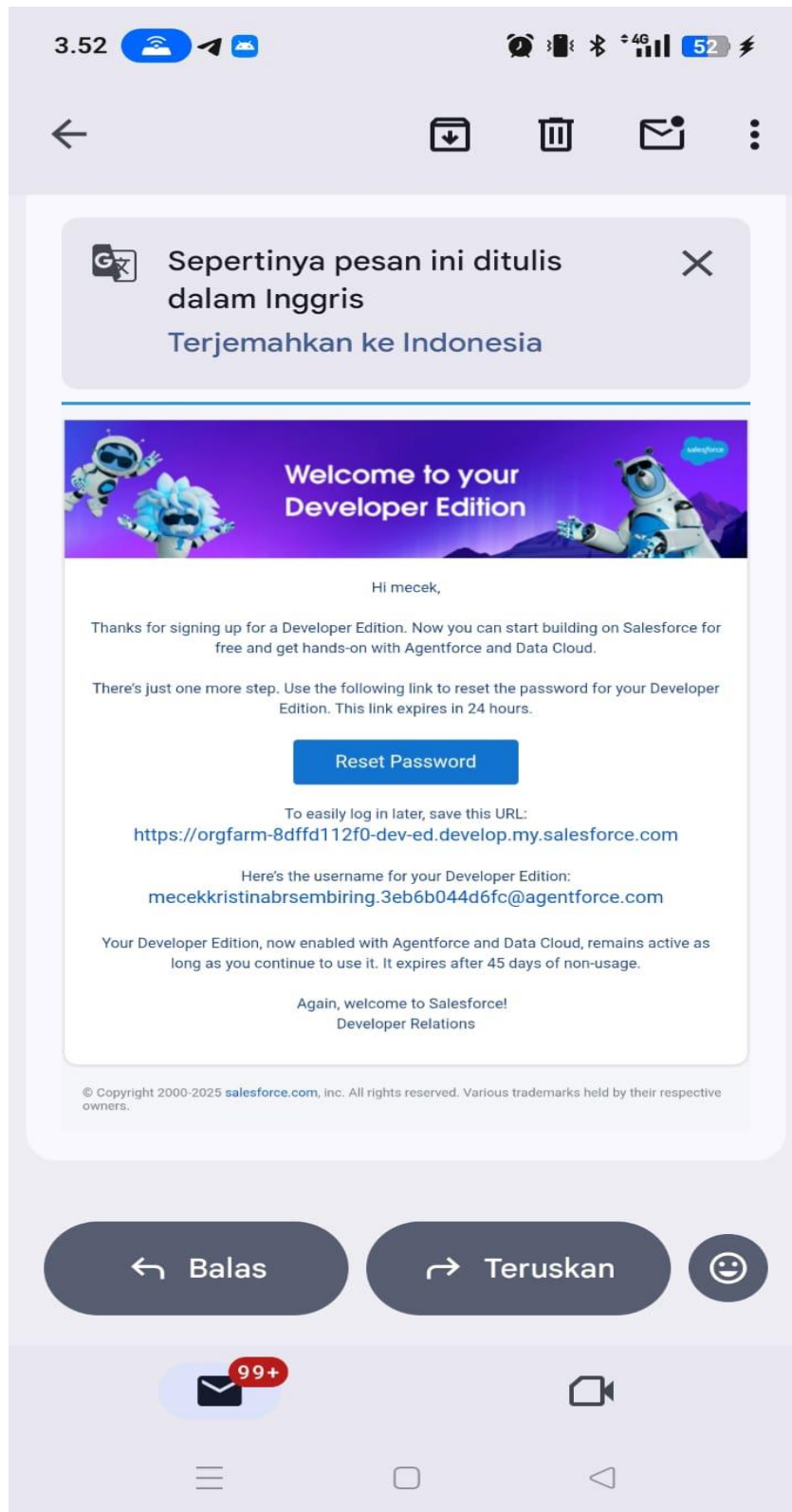
☒ I agree to the [Main Services](#)

- c. Click on **Sign me up**, then just wait until there's an email received.

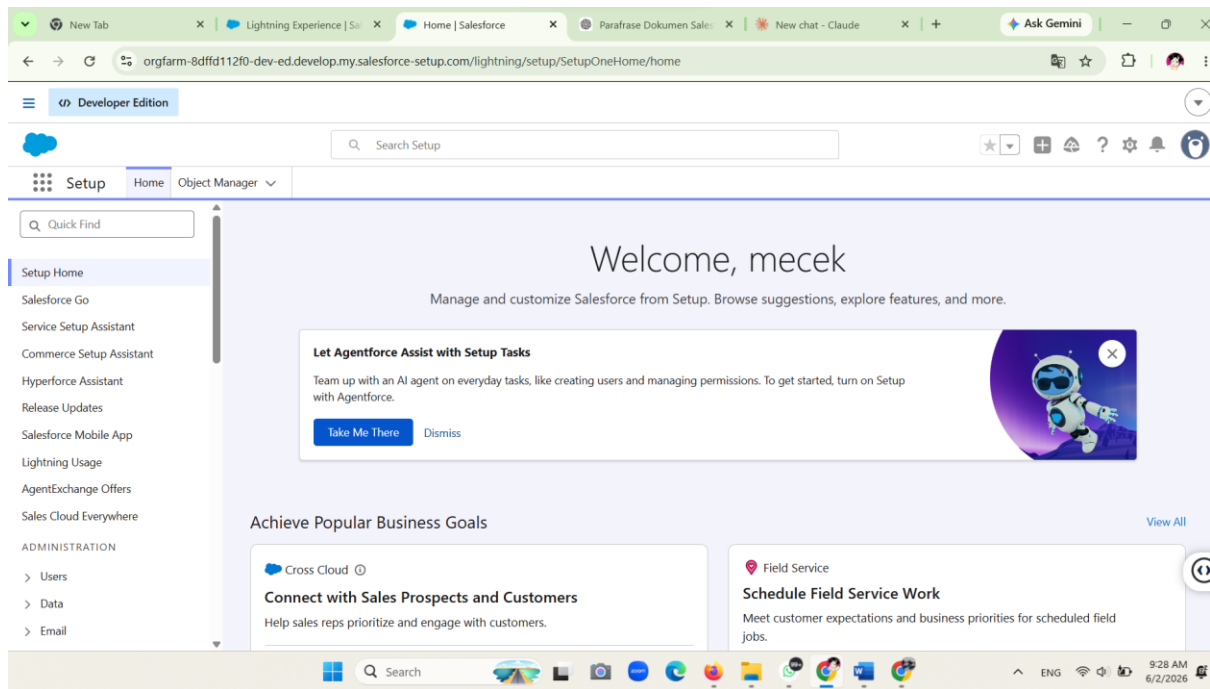


Step 2 : Account Activation

- a. Go to the inbox of the email that we used while signing up. Click on **Reset Password** to activate the account. The email may take 2-10 mins.



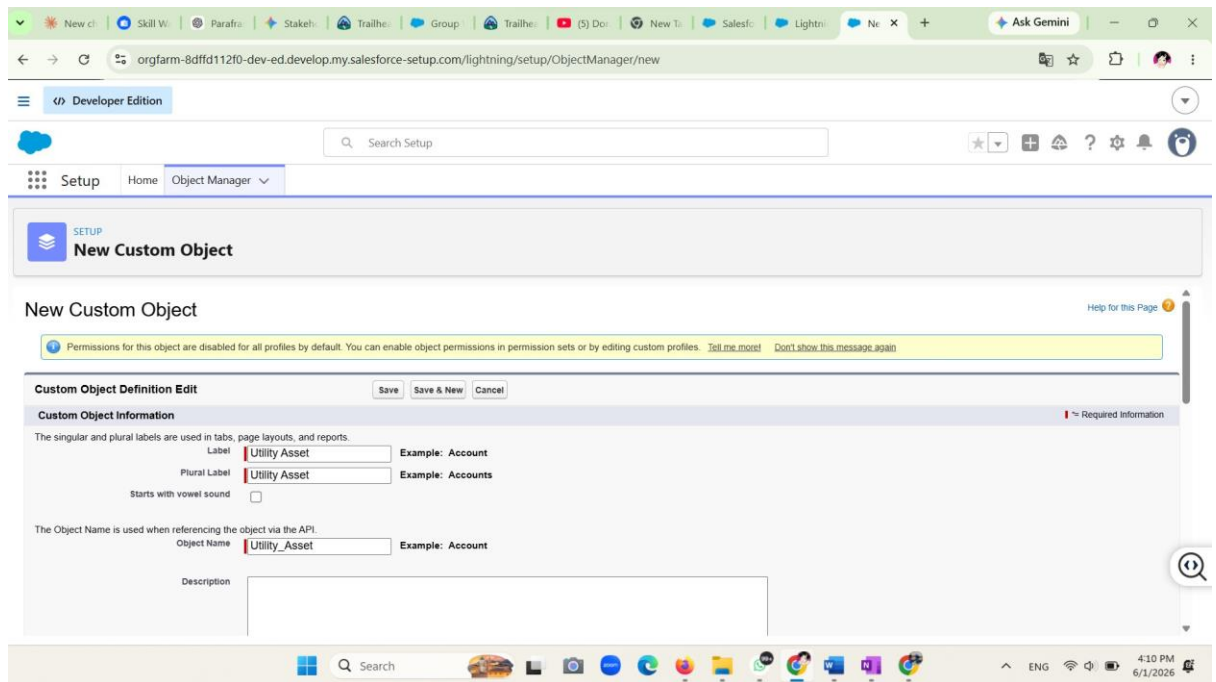
- b. Click on the **Reset Password** button.
- c. Give a password, answer a security question, and click on **Change Password**.
- d. Then we will redirect to the Salesforce Setup page.



Step 3: Custom Object Creation for Utility Asset

The core utility hardware tracking table was established by creating a new custom object within the Object Manager utility. To create an object:

1. From the setup page, click on **Object Manager**
2. Click on **Create** → Click on **Custom Object**
1. The object configuration parameters were defined as follows:
 - **Object Label:** Utility Asset
 - **Plural Label:** Utility Assets
 - **Object Name (API Name):** `Utility_Asset__c`
 - **Record Name:** Asset Name
 - **Data Type:** Auto-Number (Display Format: `MET-{0000}`, Starting Number: 1)
 - Select Allow Reports
 - Select Allow Track Field History
 - Select Allow Search
 - Click on **Save** (or **Save & New** if we want to create another Object)



Step 4: Custom Field Creation for Utility Asset

Specific technical metadata attributes were provisioned under the Fields & Relationships sub-menu of the new Utility Asset object to accommodate tracking parameters.

- **Meter ID (Meter_ID__c):** Configured as a Text field with a character length of 50. To satisfy the high-priority business requirement of sub-2-second data retrieval speeds, this field attribute was explicitly enabled as both a Unique record identifier and an Indexed External ID.
- **Asset Status (Asset_Status__c):** Configured as a Picklist data type containing explicit system values: **Active**, **Inactive**, and **Suspended**.
- **Installation Address (Installation_Address__c):** Configured as a standard Text Area field capped at 255 characters to record street location strings efficiently.
- **Customer Account (Customer_Account__c):** Established as a flexible Lookup Relationship pointing directly to the standard Account entity to tie assets to active consumers.

Step 5: Custom Object Creation for Field Visit

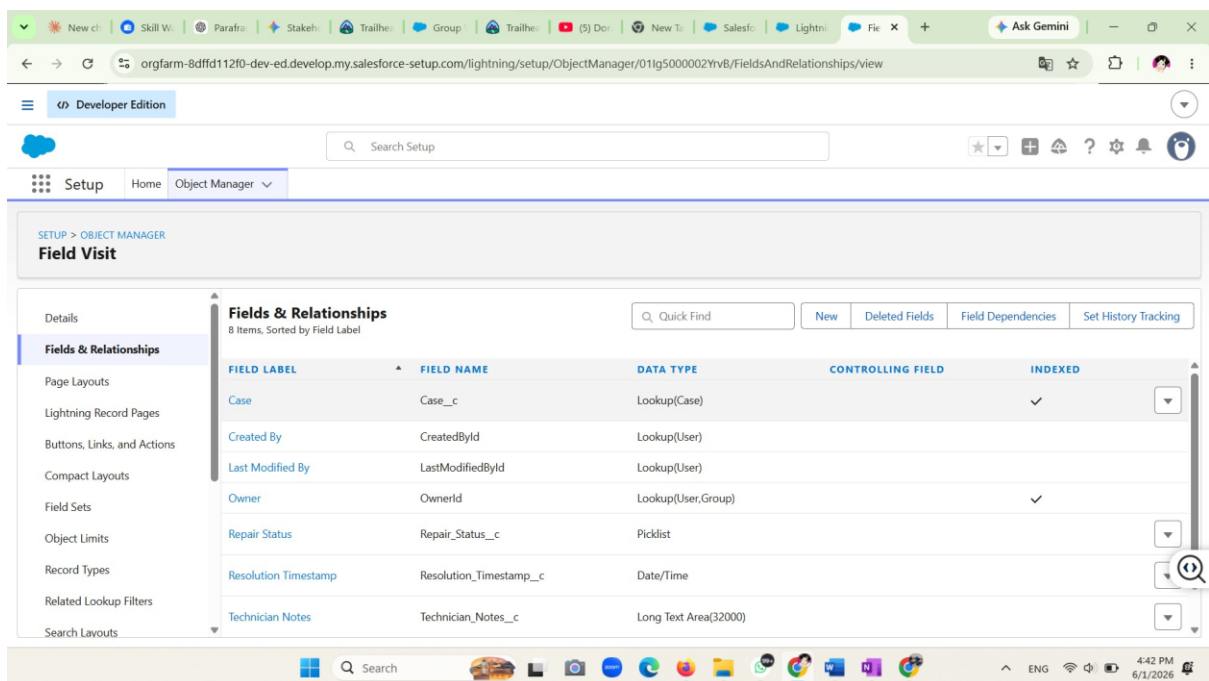
To transition the technical field crew dispatching processes away from manual paper tracking methods, a secondary custom object was generated inside the Object Manager layout as well.

- **Object Label:** Field Visit
- **Plural Label:** Field Visits
- **Object Name (API Name):** **Field_Visit__c**
- **Record Name:** Visit Number
- **Data Type:** Auto-Number (Display Format: **VIS-{0000}**, Starting Number: 1)

Follow the same steps as mentioned in **Step 3**.

Step 6: Custom Field Creation for Field Visit

To preserve operational tracking flexibility and support disconnected mobile user requirements, data properties were added to the Field Visit object framework.



The screenshot shows the Salesforce Setup interface for the 'Field_Visit__c' object. The 'Fields & Relationships' section is active, displaying a table of fields. The table has columns for FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The fields listed are Case, Created By, Last Modified By, Owner, Repair Status, Resolution Timestamp, and Technician Notes.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Case	Case__c	Lookup(Case)		✓
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Repair Status	Repair_Status__c	Picklist		
Resolution Timestamp	Resolution_Timestamp__c	Date/Time		
Technician Notes	Technician_Notes__c	Long Text Area(32000)		

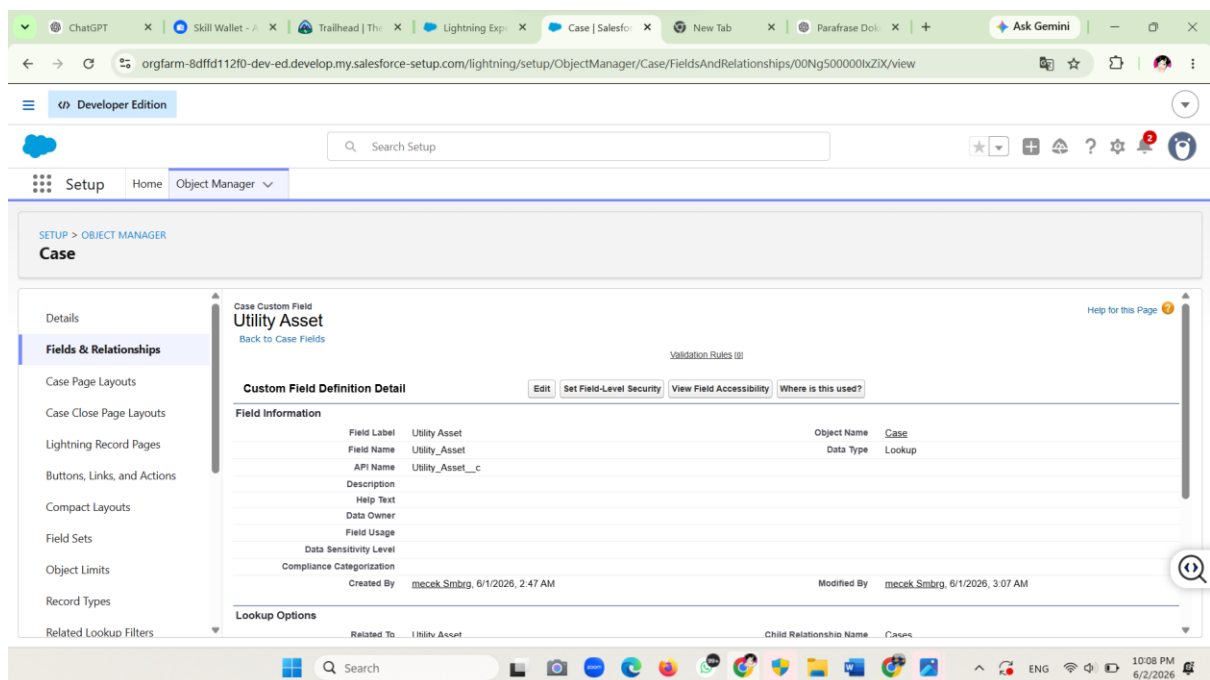
- **Parent Case (Parent_Case__c):** Configured as a standard Lookup Relationship pointing directly to the Case entity. This structural designation ensures that technical workers can maintain independent record ownership on their mobile devices while staying connected to the root support ticket.
- **Repair Status (Repair_Status__c):** Configured as a Picklist data type with workflow status values: **Assigned**, **In Progress**, **Resolved**, and **Failed**.
- **Technician Notes (Technician_Notes__c):** Configured as a Long Text Area (32,000) to allow extensive mobile technical descriptions.

- **Resolution Timestamp (**Resolution_Timestamp__c**)**: Configured as a Date/Time field to log the precise completion moment for operational efficiency metrics.

Step 7: Standard Object Metadata and Page Layout Customization (Case)

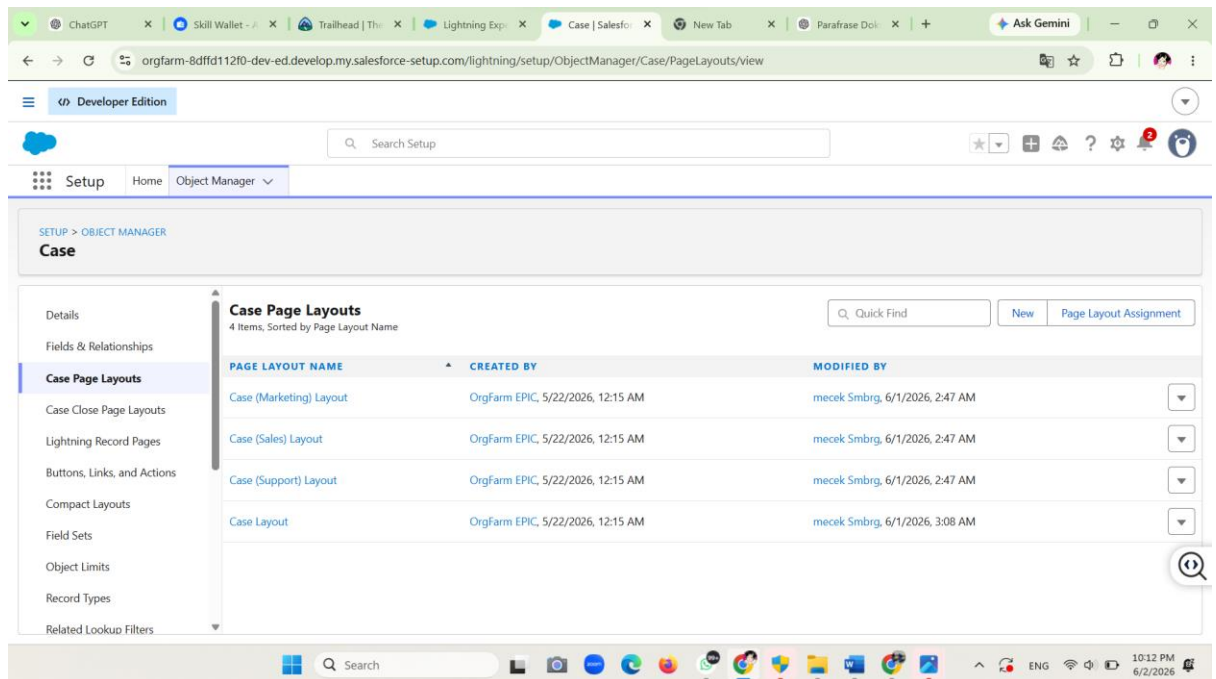
Before verifying the final system schema, the standard Case object infrastructure is customized to recognize utility-specific support categories and clean up the user interface.

- From the setup page, click on **Object Manager** in the top menu bar.
- Scroll down and click on the standard **Case** object.
- Click on **Fields & Relationships** → click the **New** button.
- Select **Lookup Relationship** as the Data Type and click **Next**.
- In the *Related To* dropdown, select **Utility Asset** and click **Next**.
- Set the **Field Label** to **Utility Asset** (the Field Name will auto-populate as **Utility_Asset**) and click **Next**.
- Click **Next** through the security visibility screens, leave the field added to the layouts, and click **Save**.

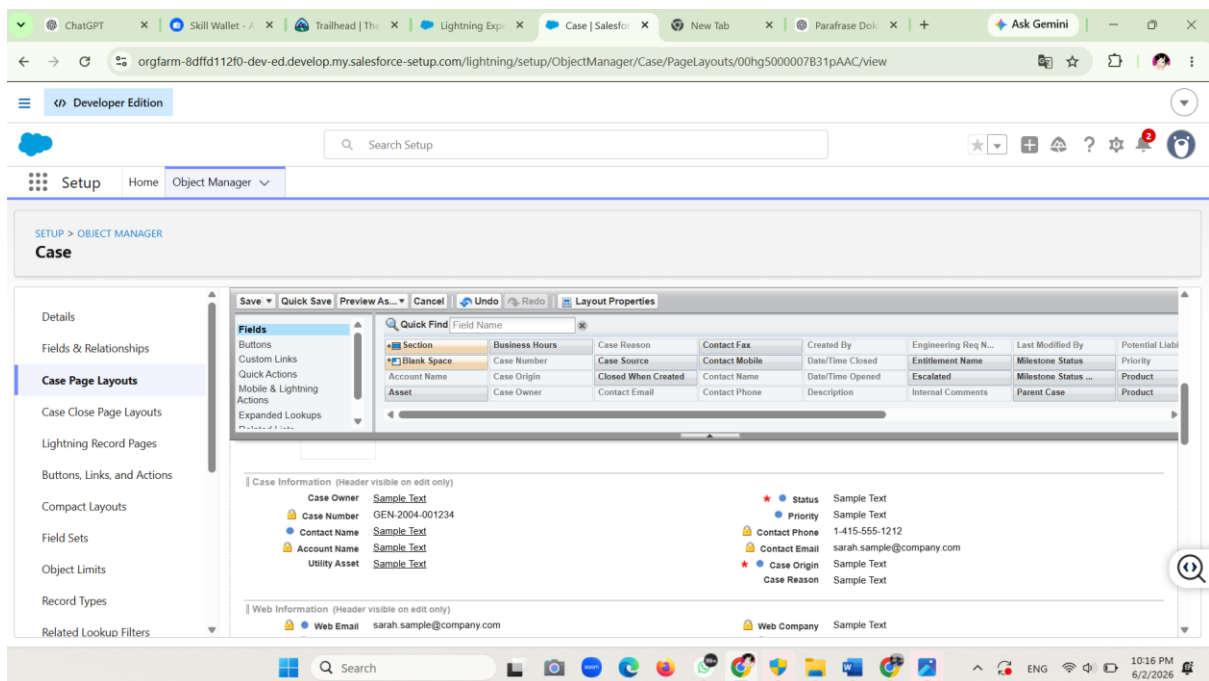


- Still under Case **Fields & Relationships**, click on the standard field **Type** (Picklist).
- Scroll to the Case Type Picklist Values related list and click the **New** button to add **Outage** and **Maintenance** on separate lines. Click **Save**.

- j. To clean up the end-user interface and enforce strict data standardization, locate the standard default values (*Mechanical, Electrical, Electronic, Structural, and Other*) within the active values table, and click **Del / Deactivate** next to each entry.
- k. From the left-hand navigation menu of the Case object, click on **Case Page Layouts** → select **Case Layout**.



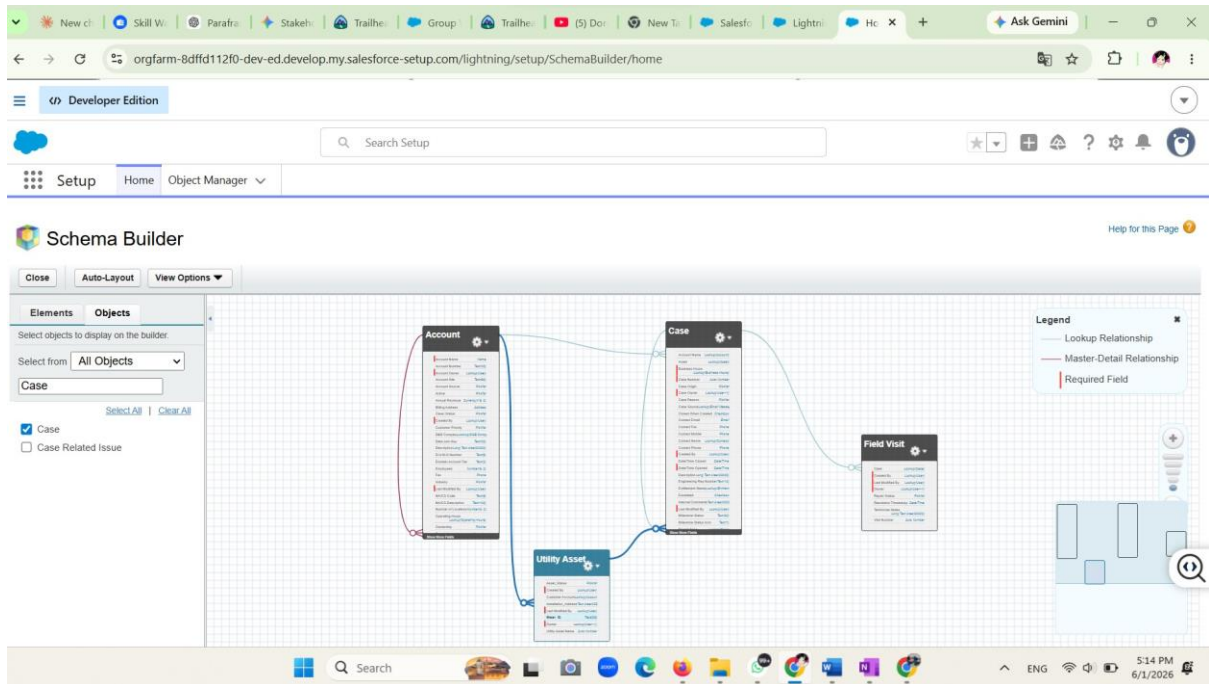
1. Streamline the Case Page Layout by removing irrelevant standard fields and positioning the custom **Utility Asset** and **Type** field directly.



m. Click **Save** on the upper left-hand layout palette menu.

Step 8: System Schema Integrity Verification

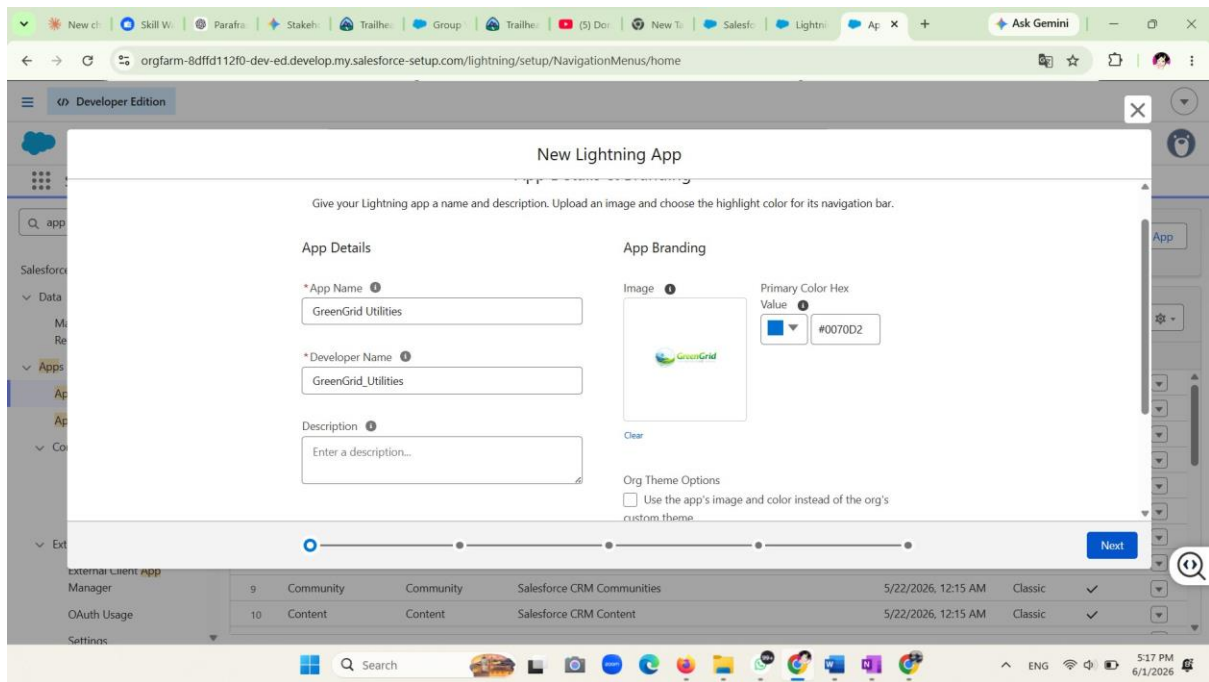
The native Salesforce Schema Builder utility was utilized to run a database layout audit. By isolating the standard Account and customized Case entities alongside the custom Utility Asset and Field Visit objects, the data model structure was visually checked to ensure reference integrity before entering the security implementation phase.



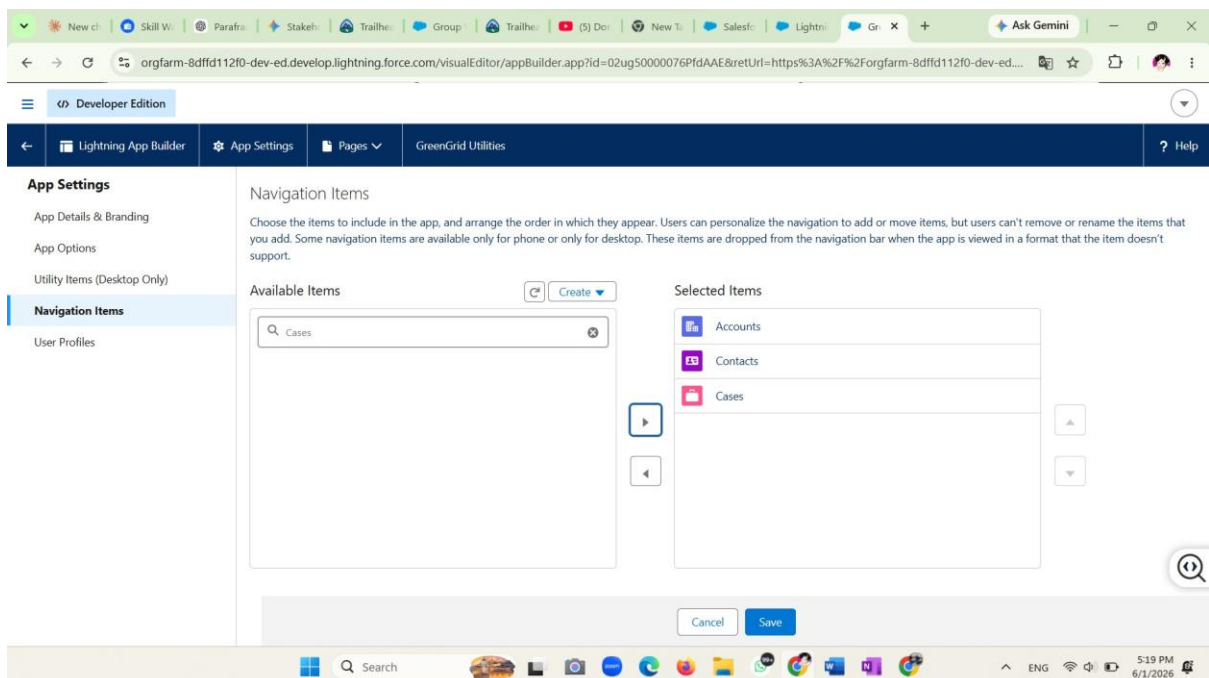
Step 9: Custom Lightning App Creation

To establish a dedicated, branded workspace layout for end-users, a custom Lightning Application container was provisioned within the platform App Manager.

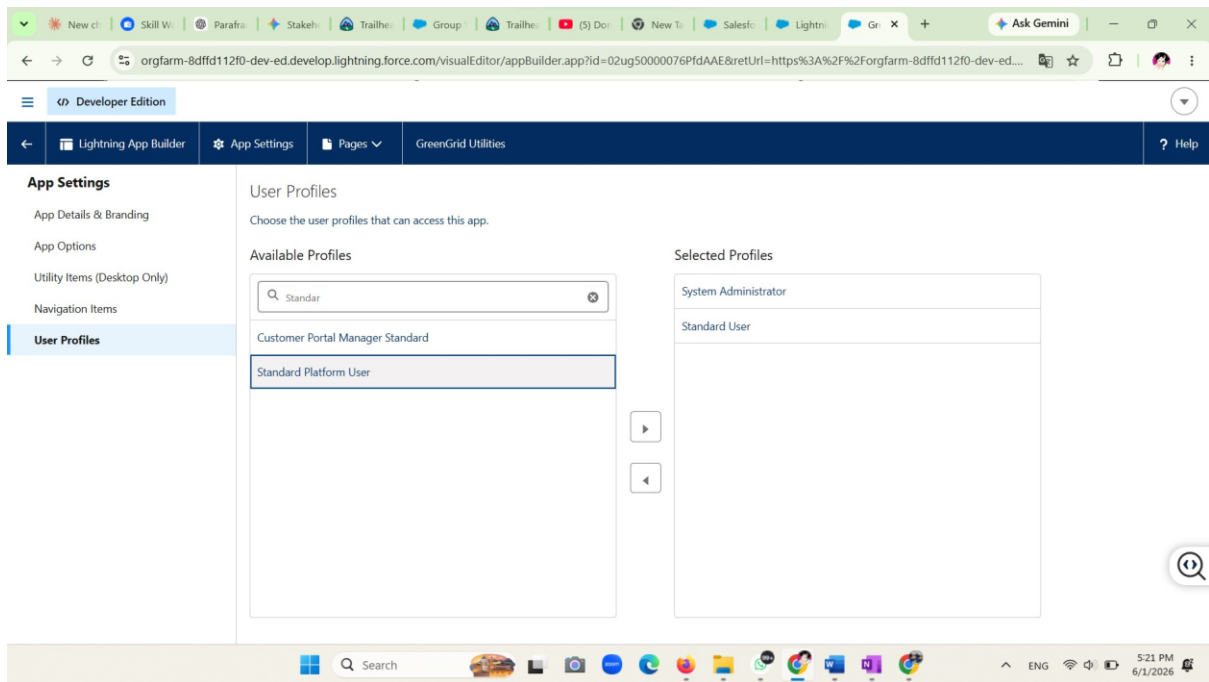
- k. Navigate to **Setup** → **App Manager** in the Quick Find bar → click on **App Manager**.
- l. Click the **New Lightning App** button located in the upper right-hand corner.
- m. App Name filled as **GreenGrid Utilities**, click **Next**.



- n. Advance through the App Options and Utility Items parameters by keeping the standard system default settings.
- o. On the **Navigation Items** panel, select standard entities **Accounts**, **Contacts**, and **Cases** from the available list and move them to the Selected Items box to form the base configuration baseline.



- p. On the **User Profiles** assignment step, scroll down and assign active visibility access to the **System Administrator** profile along with core operational user profiles.

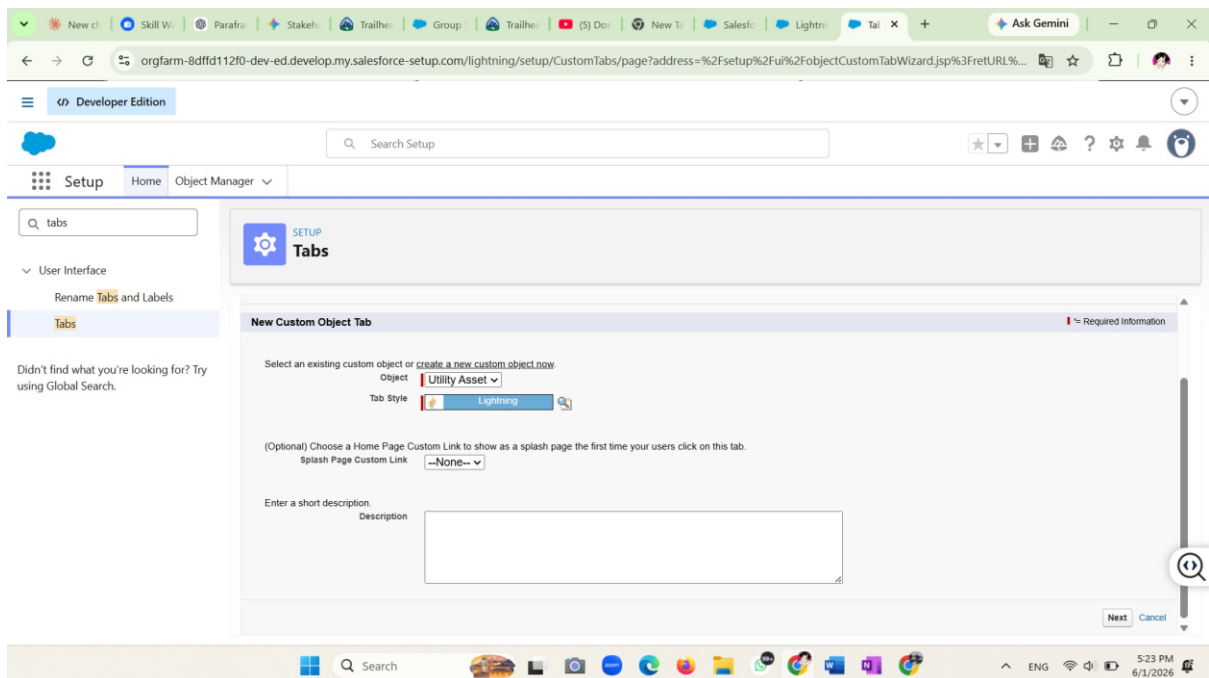


q. Click **Save & Finish**.

Step 10: Custom Object Tab Provisioning (Utility Asset)

To make the newly deployed custom tables visible and accessible to system end-users within the navigation interface, standalone user interface tabs were provisioned.

- Navigate to **Setup** → type **Tabs** in the Quick Find bar → click on **Tabs**.
- Under the **Custom Object Tabs** related list, click the **New** button.



- Click **Next**, leave the profile visibility settings at their default values, and click **Next** again.

- d. On the **Add to Custom Apps** step, uncheck the "Include Tab" global header box to remove it from irrelevant background systems, but ensure the checkbox remains explicitly checked for the **GreenGrid Utilities App**.
- e. On the **Add to Custom Apps** step, uncheck the "Include Tab" global header box to remove it from irrelevant background systems, but ensure the checkbox remains explicitly checked for the **GreenGrid Utilities App**.
- f. Ensure the checkbox for "**Append tab to users' existing personal customizations**" is checked at the bottom of the page layout.
- g. Click **Save**.

Step 11: Custom Object Tab Provisioning (Field Visit)

Following the successful deployment of the primary infrastructure layout, navigation tabs for the remaining operational objects were built just the same as **Step 9** above to complete the end-user navigation framework.

- a. Return to the **Tabs** and click **New** under the Custom Object Tabs panel.
- b. Select **Field Visit** from the Object dropdown list and assign a distinct visual Tab Style.
- c. Advance to the app routing parameters screen, exclude the tab from general background profiles, and explicitly enable it for the **GreenGrid Utilities App** navigation bar.

